

Intersection of three planes

Possibilities:

- Planes are parallel and distinct – no solution (normals are collinear)
- If only one pair of normals is collinear, then there is a pair of parallel planes
- If the normals are coplanar, but no pair is collinear, then either the planes have no point in common or they have a line in common
- If the normals are non-coplanar, then the planes intersect in a single point

Determine the intersection, if any, of each of the following sets of planes. In each case, give a geometrical interpretation of the system of equations and the solution. Also, state whether the system has no solution, a unique solution or an infinite number of solutions.

$$\pi_1: 3x - 2y + 5z = 8$$

$$1) \pi_2: 6x - 4y + 10z = -7$$

$$\pi_3: 24x - 16y + 40z = -1$$

$$\vec{n}_1 = [3, -2, 5]$$

$$\vec{n}_2 = [6, -4, 10]$$

$$\vec{n}_3 = [24, -16, 40]$$

$$\therefore \vec{n}_1 = k\vec{n}_2 = t\vec{n}_3$$

$\Rightarrow$ parallel.

$D_1 \neq kD_2 \neq tD_3 \Rightarrow$ planes are all parallel
and all distinct (no solution)

$$\pi_1: 3x + 5y - 2z - 4 = 0$$

$$2) \pi_2: 4x - 7y + 6z + 11 = 0$$

$$\pi_3: 6x + 10y - 4z + 1 = 0$$

$$\vec{n}_1 = [3, 5, -2]$$

$$\vec{n}_2 = [4, -7, 6]$$

$$\vec{n}_3 = [6, 10, -4]$$

$$* 2\vec{n}_1 = \vec{n}_3 \Rightarrow \pi_1 \text{ \& } \pi_3 \text{ are parallel}$$

$$* 2D_1 \neq D_3 \Rightarrow \pi_1 \text{ \& } \pi_3 \text{ are distinct}$$

$$* \vec{n}_2 \neq k\vec{n}_1 \neq t\vec{n}_3$$

$\Rightarrow \pi_2$ is not parallel to π_1 or π_3 .

$\therefore \pi_1$ & π_3 are parallel and distinct.

π_2 is not parallel to π_1 or π_3 .

$\Rightarrow$ no solution.

$$\pi_1: x+2y+3z-4=0$$

$$3) \pi_2: 2x-y+4z+7=0$$

$$\pi_3: 3x-14y+z-31=0$$

Done in class !

$$\pi_1: x+y-3=0$$

$$4) \pi_2: y+z+5=0$$

$$\pi_3: x+z+2=0$$

Done in class !